

developing and manufacturing liquid rocket engines for space and defence applications. Much like how Rolls-Royce supplies aircraft engines to Airbus, Thrustworks builds propulsion systems delivered to organisations developing rockets and launch vehicles.

Manan explains that the name ‘Thrustworks Dynetics’ is rooted in its core purpose. “Just like Iron Works or Coal Works—industries named after their core function—we chose Thrustworks, as we work with thrust. ‘Dynetics’ is a blend of ‘dynamics’ and ‘kinetics,’ referring to systems in motion and influenced by forces. Together, the name captures exactly what we do—working with thrust and the science of dynamic, kinetic systems.”

Thrustworks develops both rocket engine hardware and proprietary software for in-house engine design. This software is not sold commercially. By relying on internally developed tools, the team avoids dependency on foreign software and bypasses restrictive international regulations, allowing for independent creation of high-performance systems.

Their engines are intended for vehicles that reach space, such as orbital and suborbital launch vehicles, or operate at hypersonic speeds. The target customer base includes government organisations like ISRO and defence departments, as well as private space startups such as Agnikul and Skyroot.

“We simplify and accelerate the launch vehicle development process by offering engines along with the essential testing, manufacturing, and design infrastructure. One of our key differentiators is our ability to cut the typical engine development timeline from 2-3 years to just six months. This helps derisk client projects by reducing costs, saving time, and streamlining the supply chain—all under one roof,” emphasises Manan.

The startup has developed both additive (3D printing) and conventional manufacturing capabilities through a network of vendors. Testing, inspection, and post-processing are conducted in-house.

Manan notes that they have created India’s first mobile rocket engine test facility to lower the cost and time associated with engine validation.

“Although much of our manufacturing is vendor-driven, we retain critical control over design and inspection at our facility. We locally source the entire supply chain, including components, to minimise dependence on foreign imports. Currently, we operate with a lean team of six aerospace engineers, all focused on early-stage R&D and prototype testing,” explains Manan.

Thrustworks is in its pre-revenue stage and has not yet completed its first deliveries. Engine pricing is expected to range from ₹10 million to ₹20 million, with customised payment cycles based on customer development timelines. The team’s next major initiative is creating India’s first integrated rocket facility, intended to redefine how liquid rocket engines are conceptualised and manufactured domestically.

Discussing Thrustworks’ flagship engine, Anya, Manan explains that it



Tha Anya engine



(L-R) Manan Joshi, CEO & Co-Founder, Vinith Shenoy, COO & Co-Founder, Kalyani Shinde, CTO & Co-Founder, Thrustworks Dynetics

is designed for high thermal resilience and can operate in hypersonic conditions, even though thermal stress in such missions is often lower than in conventional liquid rocket applications.

The engine subsystems have passed rapid start-stop testing, making them suitable for tactical defence scenarios. Although detailed performance metrics remain confidential, Anya includes thrust vectoring capabilities and meets specific environmental qualification standards. With a thrust-to-weight ratio exceeding 110, it is both lightweight and compact, making it ideal for mobile deployment. Its fuel efficiency and impulse density are optimised for short-duration, high-speed missions, though exact figures have not been disclosed.

Despite this ambitious progress, the company faces sector-wide challenges, particularly in testing infrastructure and the availability of skilled talent. Simplified government clearances and supportive state-level policies could significantly benefit private space startups like Thrustworks. **EFY**

—Nitisha Dubey